Luis Enrique Okal Dulo

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PROFESSIONAL SUMMARY

Detail-oriented Informatics and Computer Science student at Strathmore University with hands-on experience across full-stack web and mobile development, backend systems, and applied machine learning. Builds end-to-end solutions spanning React and React Native front ends, Django REST Framework back ends, PostgreSQL databases, and scikit-learn predictive models, with additional experience integrating payment APIs (M-Pesa Daraja), embedded systems and IoT, and Microsoft Power Platform tools (Power BI, Power Apps, Power Automate). Comfortable working in Agile Scrum teams and across Linux and Windows environments, with a strong interest in building practical, data-driven software products.

WORK EXPERIENCE

Industrial Attachment, IT & Digital Transformation Division

Isuzu East Africa Ltd | January 2026 - April 2026

- Provided Local IT Help Desk support, receiving and resolving service tickets covering End User Computing issues, MS Office Suite, and internal company applications.
- Delivered hardware support and onboarding for new workstations and users, setting up laptops, desktops, and printers and integrating them onto the company network.
- Managed IT asset lifecycle tracking, including equipment movement and decommissioning, and maintained the Isuzu EA IT Managed Asset database for PCs, peripherals, phone sets, and switches.
- Supported backup and recovery processes and participated in IT projects from requirements gathering through deployment.
- Built support automation workflows using Power Apps, Power Automate, and SharePoint, and tested new systems, applications, and digital tools prior to rollout.
- Trained employees on submitting IT requests through the company's online self-help portal.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, TypeScript, Java, Kotlin, C++, SQL
- **Backend:** Django REST Framework, FastAPI, Node.js, Express, RESTful API design
- **Frontend & Mobile:** React, React Native (Expo), HTML, CSS
- **Databases:** PostgreSQL, pgvector, MongoDB, Firebase
- **Machine Learning & AI:** scikit-learn, XGBoost, Linear Regression, Isolation Forest, retrieval-augmented generation (RAG), Claude API
- **Data Engineering & BI:** ETL pipelines, star schemas, data quality checks, Power BI
- **Payments:** M-Pesa Daraja API (STK Push, callbacks), double-entry ledgers
- **Security & Testing:** OWASP Top 10 reviews, pytest, bandit, pip-audit, npm audit
- **DevOps & Automation:** Git, GitHub Actions, PowerShell, Power Apps, Power Automate, SharePoint
- **Infrastructure & IoT:** Network and switch configuration, Linux, Windows, sensor and microcontroller integration
- **Fundamentals & Methods:** Data Structures and Algorithms, Computer Networks, Computer Architecture, Agile Scrum

EDUCATION

Strathmore University

Bachelor of Science in Informatics and Computer Science

Expected Graduation: July 2027

KIPS Technical College
ICDL Certificate
Feb 2023 - May 2023

Nairobi School
Kenya Certificate of Secondary Education (KCSE)
2019 - 2022 | Grade B

PROJECTS

Smart School Feeding Management System (Capstone Project)

In partnership with Webmasters Kenya | github.com/LuisDulo/smart-school-feeding

- Platform where parents top up a child's meal balance by M-Pesa, kitchen staff serve meals by QR code, and administrators monitor balances, demand forecasts and suspicious transactions.
- Built Django REST services for accounts, meals, payments, forecasting and anomaly detection (244 backend tests), a React admin dashboard and a React Native parent app with push notifications.
- Compared Linear Regression, Random Forest and XGBoost for meal demand forecasting on 3,000 held-out rows; XGBoost reached an R^2 of 0.81 (vs 0.59) with a mean error of 1.03 meals.
- Paired an Isolation Forest with explainable rules, raising recall on injected meal-serving anomalies from 28% to 100% at 88% precision (synthetic and demo data).

Technologies Used: React Native, React.js, Django REST Framework, PostgreSQL, scikit-learn, XGBoost, M-Pesa Daraja API

Methodology: Agile Scrum (five two-week sprints)

CHAMA – Savings Group Platform

github.com/LuisDulo/chama

- Platform for running savings groups: one account runs several chamas, each with its own members, roles, versioned rules and statements; contributions arrive by M-Pesa STK Push or cash and post to a double-entry ledger.
- Credits money only when M-Pesa confirms payment and ignores repeated callbacks; met 12 of 12 acceptance criteria with 129 backend tests.

Technologies Used: Django REST Framework, PostgreSQL, React, TypeScript, M-Pesa Daraja API

Sheria RAG – Kenyan Tax, SHA and HELB Assistant

github.com/LuisDulo/sheria-rag

- Retrieval-augmented chatbot that answers only from official KRA, SHA and HELB guidance and says so when the sources do not cover a question.
- Reached Hit@12 of 0.98 on 44 labelled questions; benchmarked hybrid search against plain vector search and kept the simpler option when it did not perform better.

Technologies Used: Python, FastAPI, PostgreSQL, pgvector, Claude API, React, TypeScript

Kenya Prices Pipeline

github.com/LuisDulo/kenya-prices-pipeline

- Daily ETL pipeline that extracts EPRA pump prices and Central Bank of Kenya inflation data, loads a PostgreSQL star schema and publishes curated CSVs to a Power BI dashboard.
- Runs 8 data-quality checks before every publish, rolls back failed runs and records every rejected row with its reason.

Technologies Used: TypeScript, Node.js, PostgreSQL, GitHub Actions, PowerShell, Power BI

Smart School Feeding Security Review

github.com/LuisDulo/smart-school-feeding-security-review

- Reviewed the capstone against the OWASP Top 10 and found a critical flaw where a forged M-Pesa callback could credit a meal account without payment.
- Fixed 10 of 11 findings, each in its own pull request and proved by a test, and added CI that runs tests and security scans on every change.

Technologies Used: OWASP Top 10, pytest, pip-audit, bandit, npm audit

Artify Website

github.com/LuisDulo/ICS-Project1-

- Website where people create art on an in-browser canvas and sell it, with a marketplace, designer hiring, live chat and M-Pesa payments.

Technologies Used: Node.js, Express, MongoDB, EJS, Socket.IO, M-Pesa Daraja API

TicketMaster App

github.com/LuisDulo/TicketMasterApp

- Android event ticketing app where organisers add events and buyers pay through M-Pesa STK Push, with event data stored in Firebase.

Technologies Used: Kotlin, Jetpack Compose, Firebase, Retrofit, M-Pesa Daraja API

Proglink Website

- Web platform that connects programmers with each other and with organizations, enabling collaboration, providing learning resources, showcasing successful projects, and building a community through forums and events.

Technologies Used: HTML, JavaScript, React

ThriftMart App

- E-commerce application enabling users to buy and sell their own goods.

Technologies Used: Java, SQL

EXTRACURRICULAR ACTIVITIES

Basketball Team, Nairobi School

- Made it to the Nairobi Region Final

Model United Nations Club, Nairobi School

- Participated in the 2019 Annual Conference

Students Campaign Against Drugs Club, Nairobi School

- Member

Science Congress Club, Nairobi School

- Developed various projects as part of the club

CERTIFICATIONS

- ICDL Certificate, KIPS Technical College

HOBBIES AND INTERESTS

- Gaming
- Basketball
- Playing Pool
- Coding